WILDER GILMER ESPINOZA LUNA

Statistics | Advanced Analytics | Machine Learning

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LinkedIn GitHub HackerRank Portfolio GitLab

Professional Summary

Statistics student at the National University of San Marcos (UNMSM), focused on advanced analytics for banking: Probability of Default (PD) models, scorecards, credit approval decisions, and machine learning applied to business problems. I have developed predictive modeling and credit risk projects integrating data preparation and temporal validation, data leakage prevention, algorithm comparison, interpretability, and decision evaluation. For a risk platform, I worked with 887,000 loans and evaluated model performance and approval scenarios; I also implemented deployment workflows, automated testing, and drift monitoring. I additionally developed a probabilistic forecasting system for cases by district, using walk-forward validation and statistical significance tests. I complement this experience with business analytics: I automated reports, reducing processing time by 83
Professional focus: Advanced Analytics | Credit Risk | Databricks | Machine Learning | Banking Analytics.

Technical Skills

Credit Risk and Banking Analytics: Probability of Default (PD), credit scoring, WoE/IV scorecards, cost-based threshold analysis, backtesting, provisions and loan loss provisioning, KPIs.
Modeling and Statistics: Regression, classification, forecasting (quantile and time-series), feature engineering, out-of-time validation, data leakage prevention, ROC-AUC, KS, Gini, calibration, DeLong tests, Diebold–Mariano tests and Holm correction, SHAP.
Machine Learning: scikit-learn, CatBoost, XGBoost, LightGBM, Optuna, PyTorch (Transformer implemented from scratch).
Data and Cloud: Python, PySpark, SQL, R, pandas, ETL/ELT; Databricks (Jobs, Delta Lake, Unity Catalog, MLflow, Model Serving; hands-on experience), Google Cloud Run, Azure (Container Apps, Azure OpenAI, Terraform; fundamentals), AWS (basic knowledge).
MLOps and Engineering: CI/CD, champion/challenger, A/B testing, drift monitoring (PSI, KS), FastAPI, Docker, GitHub Actions, automated testing, CodeQL, Streamlit.
Generative AI and Other Tools: LLMs, RAG, embeddings, AI agents, MCP, Power BI, QGIS, Scrum, Jira.

Professional Experience

Data Analytics and Business Intelligence | Mebol 2024 – 2025 (6 months)
Lima, Peru

- Reduced inventory record errors by 28% using SQL queries to integrate, clean, and validate data from multiple sources, improving consistency and integrity.
- Automated reporting processes with SQL and Excel, reducing processing time from approximately 6 hours to 1 hour per week (83%) and freeing up 5 work hours weekly.
- Used Python for exploratory data analysis (EDA), identifying anomalies, outliers, variations, and trends, helping detect operational inconsistencies and patterns.
- Used PySpark to process and transform large volumes of data in a distributed environment, preparing information for analytics and analytical workflows.
- Defined validation rules and standardization criteria with the warehouse, purchasing, and sales teams, strengthening data quality and reliability.
- Built Power BI dashboards to monitor the availability and traceability of more than 450 products, supporting KPI tracking and operational decision-making.

Selected Projects

Credit Risk Platform | Predictive Modeling and MLOps Repository Demo
Python, CatBoost, XGBoost, LightGBM, Optuna, SHAP, Databricks, Delta Lake, MLflow, FastAPI, Streamlit, GitHub Actions

- Built a Probability of Default (PD) model and a credit risk scorecard using 887,000 Lending Club loans with known outcomes from a dataset of 2.26 million loans. Applied out-of-time temporal validation, excluded 22 post-origination variables to prevent data leakage, and compared four algorithms. The model achieved a ROC-AUC of 0.691, Gini of 0.382, and KS of 0.277 on the test set. Benchmarked it against a traditional WoE/IV scorecard (ROC-AUC of 0.681), used as a regulatory benchmark.
- Evaluated credit policy scenarios using a decision threshold based on the relative costs of false negatives and false positives (5:1). At a cutoff of 0.18, the scenario approves 61.3% of applications, identifies 61.9% of defaults, and reduces the default rate in the approved portfolio from 15.5% to 9.6%.
- Implemented a Level 2 MLOps workflow on Databricks (Jobs, Delta Lake, Unity Catalog, MLflow, and Model Serving), with CI/CD across development, staging, and production, model registration and deployment, A/B testing using a champion/challenger strategy, and rollback. Added monitoring with PSI, KS, DDM, and Page-Hinkley; during a temporal replay, the system detected data drift and concept drift and triggered two retraining runs. The challenger was not promoted due to insufficient statistical evidence ($p = 0.61$).
- Added 169 automated tests with 100% coverage and static analysis using CodeQL to improve code reliability.

GeoBanca Norte Peru | Spatial Analytics for Financial Channels Repository Interactive Poster
Python, GeoPandas, PySAL, spatial models, Kriging, Random Forest, SHAP, integer optimization, QGIS

- Integrated data from SBS, INEI, SENAMHI, and OpenStreetMap by UBIGEO for 327 districts in northern Peru (7.3 million residents and 629 branches), and assessed locations for new banking agents, ATMs, and special-purpose branches (EOB), accounting for opportunity and climate risk.
- Identified spatial dependence in access to financial channels (Moran's I = 0.54, LISA) and compared OLS, SAR, SEM, and spatial

Durbin models. The SDM achieved the lowest AIC (1648 versus 1696 for OLS) and eliminated residual autocorrelation.
- Modeled extreme rainfall associated with El Niño (1998, 2017, and 2023) using semivariograms and kriging, reducing cross-validation RMSE by 23% compared with IDW (72.5 versus 94.7 mm).
- Ranked districts by opportunity and risk and optimized locations using integer programming (MCLP, CBC solver), covering 63,643 adults with 25 sites. Published the results in an interactive poster with maps and 29 figures.

RiskRAG | Generative AI for Credit Risk and Insurance Regulations Repository Demo

Python, NLP, RAG, vector search, BM25, FastAPI, PostgreSQL, pgvector, Azure (Terraform), Docker, MCP

- Designed a hybrid RAG system combining vector search and BM25, hierarchical regulatory-document chunking, Reciprocal Rank Fusion (RRF), and reranking to retrieve regulatory information.
- Implemented automated citation verification, abstention mechanisms, and deterministic tools based on YAML rules and Decimal arithmetic to classify debtors and calculate provisions under SBS regulations.
- Developed MCP servers with permission separation to integrate regulatory queries, financial calculations, and Probability of Default inference (default model exposed as an API) with AI agents.
- Implemented automated evaluation using 8 metrics and CI/CD quality controls: 89.6% citation precision, 91.7% intent detection, and 100% adversarial containment on the evaluated dataset.
- Added LLM security controls, prompt-injection detection, and personal-data validation; automated 88 tests and DevSecOps analysis with GitHub Actions, Trivy, Gitleaks, and Checkov. The Azure infrastructure is defined in Terraform and validated, but has not yet been deployed to a real subscription.

Hackathons and Competitions

Rumbo | Multi-Agent Platform for Matching Candidates and Jobs Project Video

All Things Agentic Hackathon 2026, Taskmaster Track (Google Cloud) | Team of 4

- Contributed to building a platform that normalizes job roles and vacancy requirements, evaluates candidate-job fit, and generates learning paths to close skills gaps.
- Participated in designing three agents using Gemini and Google ADK (Role Classifier, Requirement Extractor, and Fit Auditor), as well as the statistical matching score.
- Contributed to a three-stage pipeline: exact matching, preselection using embeddings and vector search in Firestore, and LLM reasoning through a batched call to reduce inference calls and costs.
- Contributed to an architecture with two deterministic Python workflows and a state machine for progressive information disclosure with explicit consent. The application was deployed on Cloud Run using FastAPI and React.

Orchestrate | OpenAI × HackerRank Event Repository

Ranked 54th globally out of 60,000+ participants (Score: 71.5/100) | 24-hour hackathon | October 24, 2026

- Designed a data-deletion and privacy audit agent (GDPR Art. 17) to detect inconsistencies and residual data ("ghost data") across databases, replicas, logs, and screenshots in 518 complex requests.
- Implemented the architecture around the principle "the model perceives; deterministic code decides": integrated Gemini 3.8 Flash for structured vision with JSON schemas and SHA-256 caching, achieving 100% reproducibility and zero API calls in production.
- Developed a multivariable heuristic attribution engine (time windows, sub-minute signatures modulo 60 seconds, and log-likelihood scoring with a margin of ≥ 2.0) to link asynchronous worker events without visible identifiers.
- Achieved 28.2/30 (94%) data accuracy (output CSV), perfect 1.000 accuracy in remediation steps, 32/32 correct states in benchmarks, and zero contractual-invariant violations in prompt-injection tests.

Education

Statistics Studies | National University of San Marcos (UNMSM), Lima 2022 – 2027

- Coursework in statistical modeling, quantitative analysis, and data interpretation for evidence-based decision-making.
- PRONABEC scholarship recipient (2024) for outstanding academic performance.

GCI World 2026 | Matsuo-Iwasawa Laboratory, The University of Tokyo Sep 2026 – Present

International Data Science and Machine Learning Program

- Learned to transform data into actionable insights and business opportunities.
- Developed skills in supervised modeling and time-series analysis, applied to marketing, customer behavior, and strategic decision-making.

Certifications

Advanced Machine Learning Operations (Databricks) | MLOps: Machine Learning Operations Specialization (Duke University) | Machine Learning Specialization (DeepLearning.AI and Stanford University)

Languages

Spanish: Native | **English:** Advanced English as a Second Language, Saylor University